

CITY UNIVERSITY OF HONG KONG

Course code & title : CS5489 Machine Learning: Algorithms & Applications

Session : Midterm, Question Samples

Time allowed : N/A

This question paper has 4 pages (including this cover page).

1. This sample paper consists of 4 questions.
 2. Answer ALL questions.
 3. Write your answers in this question paper.
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*This is a **closed-book** examination.*

Candidates are allowed to use the following materials/aids:

One A4 page (single-sided only) of handwritten notes with physical pen or pencil. Digital version or print of digital version is not allowed.

Materials/aids other than those stated above are not permitted. Candidates will be subject to disciplinary action if any unauthorized materials or aids are found on them.

Student EID: _____

Student ID: _____

Seat Number: _____

Multiple Choice/Selection Questions (10 points)

5 marks each. *For a multiple selection question, an incorrect answer will be penalized $5/K$ marks, where K is the number of correct answers. If more incorrect answers are given than correct answers, the marks will be 0.*

Q1. Which statements about generative classification model are correct? (select all that apply)

- A) They estimate probability distributions of features from each class.
- B) It is hard to add prior knowledge to the classifier.
- C) It can only work with binary classification problems.
- D) Selecting different probability distributions can obtain different classification performance.
- E) Naïve Bayes classifiers assume each feature dimension is modeled independently.

Answer: <A, D, E>

Q2. Which of the following statements about Support Vector Machines (SVM) are correct?

- A) The objective of SVM is to minimize the margin between the separating hyperplane and the nearest data points.
- B) In soft-margin SVM, slack variables are introduced to allow some misclassification.
- C) The margin on one side of the boundary could be larger than that of the other side.
- D) The kernel trick allows SVM to classify non-linearly separable data.
- E) Kernel methods can transform non-linear problem into a more efficient problem in the feature space.

Answer: <B, D>

Discussion Questions (20 marks)

10 marks each question.

Q7. You are working for a “smart pen” company, which can scan text using a handheld pen device. Your job is to build a model to classify the text the pen reads into different sentiment categories, such as “happy”, “sad”, “neutral”, which will then turn on different lights on the pen as a special effect. Because of the limited processing power on the pen, you will use a generative model classifier with bag-of-words model. A training set is provided, consisting of 5000 sentences and their class labels. Your initial classifier is Naïve Bayes Bernoulli model, but the performance is poor. How could you improve the model performance? (List 4 improvements)

[Any 4, +2.5 points each. -2.5 points if a reason is not correct]

- 1) We can try other more appropriate probability distributions, like multinomial.
- 2) Using n-gram bag-of-words
- 3) Perform text preprocessing (stemming, lemmatisation, and removing numbers or punctuation, etc.) before bag-of-words.
- 4) adjust the Laplace smoothing hyperparameter on the probability distributions.
- 5) adjust the vocabulary length.
- 6) try different representations, e.g., TF, TF-IDF.

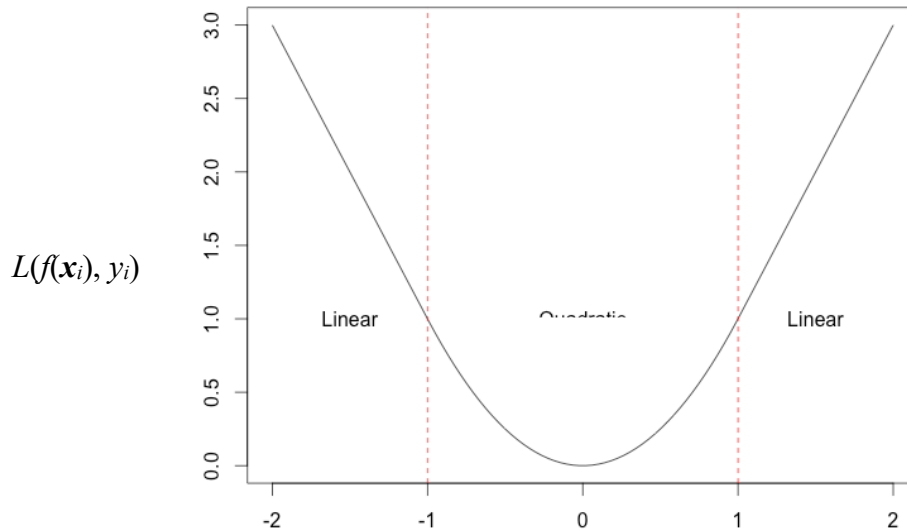
Using word vector representation is a not-quite correct answer, but it will incur heavy processing -- we need to store the vector representations for all tokens, then we need to perform vector-vector multiplication for a linear classifier. Can give +1.5 point for it.

Q4. Consider linear regression using the Huber loss function:

$$L(f(\mathbf{x}_i), y_i) = \begin{cases} \frac{1}{2}(y_i - f(\mathbf{x}_i))^2, & \text{if } |y_i - f(\mathbf{x}_i)| < 1 \\ |y_i - f(\mathbf{x}_i)| - \frac{1}{2}, & \text{otherwise} \end{cases}$$

(a) Draw a plot of the loss function.

(a) [5 marks]



(b) Use chain rule to calculate the derivative: Consider a simple neural network with one hidden layer. The network makes a prediction $\hat{y}$ as follows:

- **Input:** $x = 2.0$
- **Hidden layer:** $h = \tanh(w_1 x + b_1)$ where $w_1 = 0.5$, $b_1 = 0.1$
- **Output layer:** $\hat{y} = w_2 h + b_2$ where $w_2 = -1.2$, $b_2 = 0.3$
- **Loss function:** $L = (y - \hat{y})^2$ where $y = 0.8$ (true value)

Using the chain rule, derive the expression for the gradient of the loss L with respect to the weight parameter w_1 and then update the weight accordingly (you may need to define variables by yourself, you are not supposed to calculate the concrete value).

To find $\partial L / \partial w_1$ via the chain rule, we need $\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial w_1}$, where

- $\frac{\partial L}{\partial \hat{y}} = (-1) \times 2 \times (y - \hat{y})$
- $\frac{\partial \hat{y}}{\partial h} = w_2$
- $z = w_1 x + b_1$ and $\frac{\partial h}{\partial z} = \text{sech}^2 z$
- $\frac{\partial z}{\partial w_1} = x$

The update can be achieved via $w_1' \leftarrow w_1 - \eta \frac{\partial L}{\partial w_1}$ where η is the learning rate.

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